

Albert Zhang

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Experience

Insource Services

May 2025 – Present

Data Solutions Engineer

Wellesley, MA

- First-authored TutorMoments-Preview with collaborators at Ai2, MIT, and Stanford, building the replay harness that scored 7,280 model tutoring decisions across 7 models against a human baseline, and shipping the public leaderboard, evaluation code, and dataset.
- Built and ran the evaluation pipeline on AWS benchmarking 11 vision-language models on 56,000+ multimodal QA pairs over handwritten student work, scoring with BERTScore and LLM-as-a-judge rubrics.
- Designed, built, and deployed to production a lead-discovery system on Azure (Django, Terraform, GitHub Actions CI/CD) that analyzes 14,000+ investment firms, with a Celery/Redis pipeline turning raw regulatory filings into ranked intelligence reports via LLM APIs.
- Architected an ELT pipeline in Snowflake, dbt, and Dagster, ingesting nested GraphQL API responses into landing tables and flattening them through layered dbt models into queryable marts, orchestrated end to end with event-triggered transformations.

Aramco

May 2024 – Aug. 2024

Research and Development Intern

Boston, MA

- Optimized a bio-polymer aerogel for carbon capture through iterative synthesis and testing, improving CO₂ capture efficiency 96% over baseline, and first-authored the resulting paper.

Research

TutorMoments-Preview: When Help is Unhelpful — Evaluating AI Tutors for Productive Struggle

A. Zhang, A. Ross, K. Patel, J. Bernado, R. Bowie, A. Trindade Ribeiro, D. Halper, H. Valayaputtur, J. Andreas, S. Loeb, L. Li, K. Lo, R. Knight. Preprint, 2026. [paper](#) · [site](#) · [code](#) · [data](#)

The Aftermath of DrawEduMath: Vision Language Models Underperform with Struggling Students and Misdiagnose Errors

L. Li, **A. Zhang**, N. Anderson, R. Knight, K. Lo. Preprint, 2026. Extends DrawEduMath (NAACL 2025 Outstanding Paper). [arXiv](#) · [site](#)

Projects

Mandate | Python, FastAPI, Supabase, ARQ, Docker

2026

- Built an RIA prospecting platform that scores the full universe of 16,600+ SEC-registered advisers against a fund's stated criteria and returns a ranked list with generated fit briefs.
- Offloaded scoring to an ARQ/Redis worker split from the FastAPI service so full-universe runs execute in the background, over Supabase Postgres with pytest coverage and Docker Compose deployment.

Compliance Analyzer | TypeScript, Next.js, Claude API, Docker

2026

- Built a document-audit system that checks internal policies against a regulatory source, sustaining 10-way concurrent scoring under API rate limits with prompt caching over the policy corpus.
- Batched corpora exceeding the model's context window and scored every requirement match with an evidence quote and a confidence level for reviewer override.

Blackbox | Python, Qdrant, Ollama, Docker

github.com/Bdfihn/Blackbox

- Built a self-hosted logging system that indexes PC activity, git history, and wearable data in a local Qdrant store and answers natural-language questions over it, running fully offline on Ollama.

Education

Boston University

Sep. 2021 – May 2025

B.A. Computer Science and B.A. Biochemistry & Molecular Biology

Boston, MA

Technical Skills

Languages: Python, SQL, TypeScript, Java, C/C++

ML & Evaluation: LLM/VLM evaluation (benchmark design, LLM-as-a-judge, BERTScore, rubric design), large-scale batch inference, RAG and vector search, LLM/VLM APIs (Anthropic, OpenAI, Google), scikit-learn

Backend & Data: FastAPI, Django, Celery, ARQ, PostgreSQL, Supabase, Qdrant, Redis, Snowflake, dbt, Dagster

Cloud & DevOps: AWS, Azure, GCP, Docker, Terraform, GitHub Actions, pytest